

SPECIAL STEELS

STOCKHOLDING & MANUFACTURING

GRADE 1.4034 - 420 STAINLESS STEEL

1.4034 is a martensitic grade of 420 stainless steel. It is magnetic and hardenable with rust resistant properties. 1.4034 stainless is characterised by having a 0.43-0.50 carbon, and 12.5-14.5% chromium content. This mix of analysis gives our 1.4034 stainless sheet and sections an outstanding corrosion resistance and exceptional wear properties after heat treatment. Superior to 410 and 416 type stainlesses for most applications. 1.4034 stainless steel has good ductility in the annealed condition, but is capable of achieving 57HRC after treatment. Optimal corrosion resistance is realised when 1.4034 is surface ground and polished after hardening.

1.4034 - 420 Stainless Steel Chemical Analysis wt %						
C	Si	Mn	P	S	Cr	Ni
0.43-0.50	≤ 1.00	≤ 1.00	≤ 0.04	≤ 0.015	12.5-14.5	≤ 1.00

APPLICATIONS FOR 1.4034

Dental and surgical instruments cutlery pump shafts gear pinions, cams, various hand tools, straight edges, parts for firearms and scissors

Not recommended for high pressure vessels.

SOFT ANNEALING 1.4034

Heat to 730-780°C. Cool slowly in air.

Structure is ferrite with spherical carbides.

MACHINEABILITY 1.4034

Good machining characteristics due to the addition of Sulphur.

High speeds and feeds producing short brittle chips

1.4034 - 420 Stainless Mechanical Properties at ambient temperature
Condition annealed. Quenched and tempered 850°C
0.2% Proof Stress: 650 Nmm ²
Tensile Strength: 850 - 1000 Nmm ²
Elongation: 10%
Hardness HB Max 230

FORGING 1.4034

Hot forming temperature 800-1100°C. Slow cool

HARDENING 1.4034

Harden from a temperature 980-1030°C

Follow by oil or air quenching.

Hardness after quenching 52-58 HRC

TEMPERING 1.4034

Tempering temperature 100-200°C

HEAT TREATMENT ADVISE OF 1.4034

Treatment temperatures, including the rate of heating, cooling and soaking times will vary due to factors such as the shape and size of each 420 stainless steel component. other considerations during the heat treatment process include the type of furnace, quenching medium and work piece transfer facilities. Please consult your heat treatment provider for full guidance for the heat treatment of 1.4034

CORROSION RESISTANCE 1.4034

Hardened condition realises resistance to atmosphere, fresh, dilute acids and alkalise, fresh water and fruit and vegetable juices

1.4034 - 420 Stainless Steel Physical properties
Average values at ambient temperature
Modulus of elasticity. ($10^3 \times \text{N/mm}^2$) : 220
Density. (g/cm^3): 7.74
Thermal conductivity. (W/m.K): 30.0
Electric resistivity. ($\text{Ohm mm}^2/\text{m}$): 0.65
Specific heat capacity. (J/g.K): 0.46
Magnetisable. Yes

WELDING OF 1.4034

Not commonly welded due to its air hardening characteristics. Welding may possibly be performed after preheating to 150-205°C with the post welding tempering at temperature for 2 hours. Recommendation of specialist advice to be sought

COLD WORKING OF 1.4034

Material will withstand only minor cold work. Radical forming operations will result in cracking

Disclaimer

The information and data presented herein are typical or average values and are not a guarantee of maximum or minimum values. Applications specifically suggested for material described herein are made solely for the purpose of illustrations to enable the reader to make their own evaluations. They are not intended to either express or imply the suitability for these or any other purposes

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